Experiment 10 - INTERNETWORKING

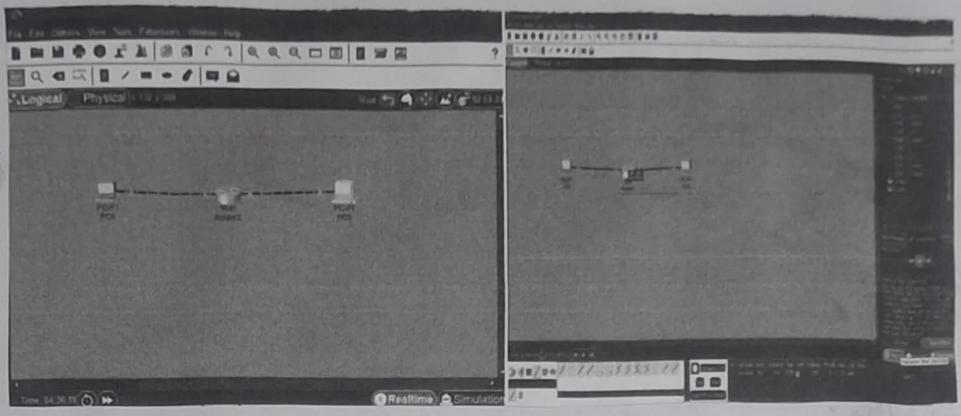
10a) Internetworking using Router

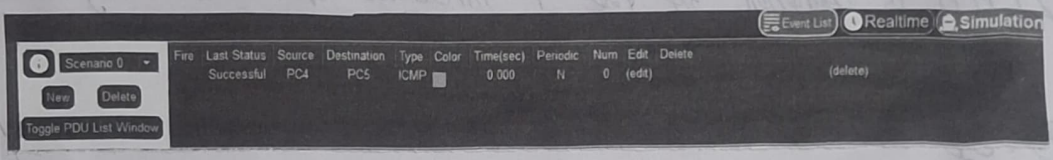
Aim:-

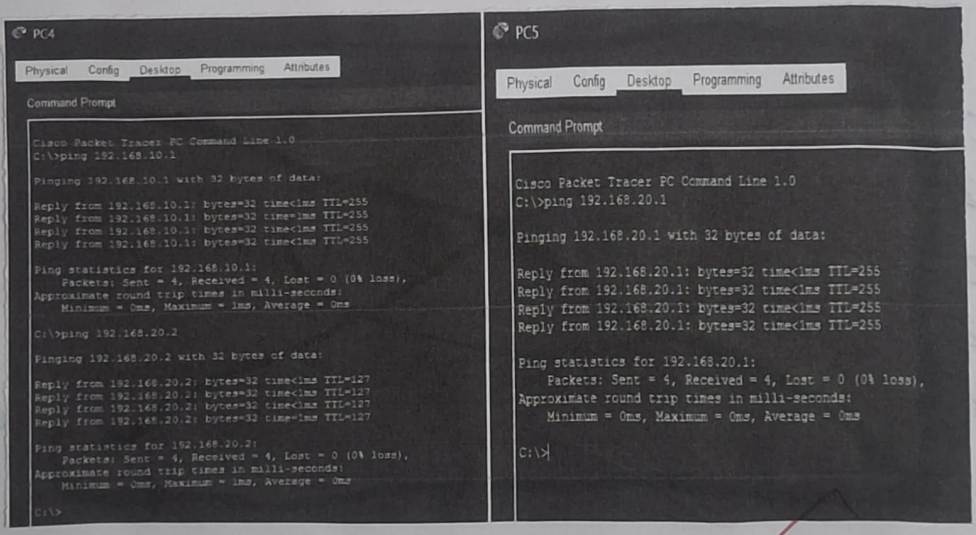
To design and Configure a simple internetwork using a router in Cisco Packet Tracer and verify Connectivity between two PCs.

Observation:-

1.) A router and two PCs are used to create a simple internetwork.
2.) PC0 is connected to the 1st interface of Router1 using a Copper Straight-through Cable.
3.) PC1 is connected to the second interface of Router1.
4.) Router interfaces are assigned different network addresses
   - 192.168.10.1/24
   - 192.168.20.1/24
5.) PC0 is assigned IP address 192.168.20.2 with gateway 192.168.20.1
6.) PC1 is assigned IP address 192.168.20.2 with gateway 192.168.20.1
7.) The router forwards packets between two networks
8.) A PDU is sent from PC0 to PC1 to verify network connectivity


Logical
Physical
Realtime
Simulation


Event List
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Successful
PC4
PC5
ICMP
0.000
N
0
(edit)
(delete)


PC4
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
PC5
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.1
Pinging 192.168.20.1 with 32 bytes of data:
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.20.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>

Result:-

Thus, a simple internetwork was successfully designed and configured using a router, and communication between PC0 and PC1 was successfully verified

10)

b. Wireless Router, DHCP Server and Internet cloud

Aim:-

To design and Configure an internetwork using a wireless router, DHCP server and Internet cloud

Observation:-

1.) A wireless router, PC, Laptop, Cable modem, INternet cloud and Cisco.com server are connected.

2.) The wireless router is configured with the SSID Home network

3.) DHCP is enabled on the wireless router

4.) The PC obtains its IPv4 address automatically through DHCP

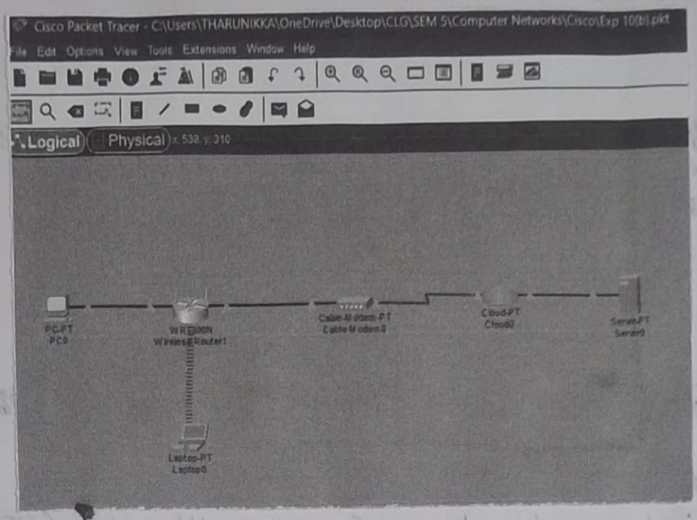
5.) The laptop is configured with a wireless module and connected to homenetwork

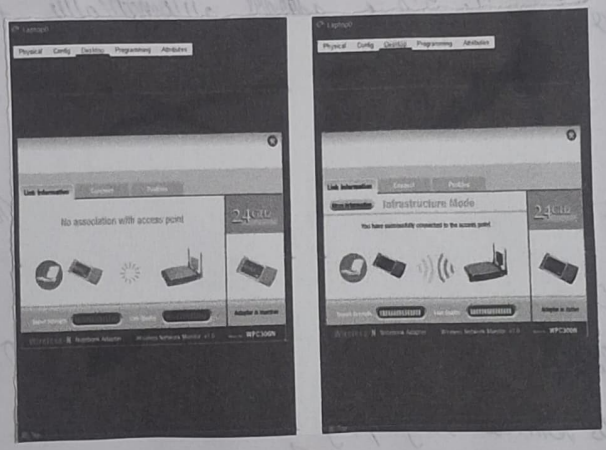
6.) The internet cloud is connected to the cable modem

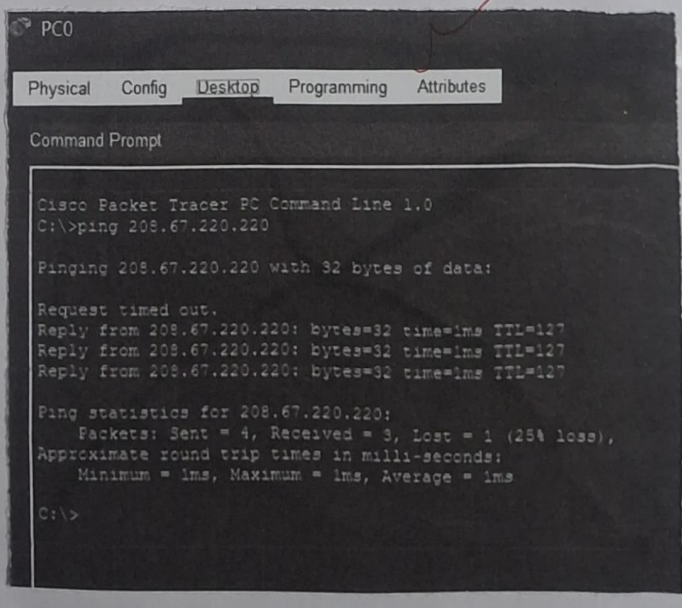
7.) The Cisco.com server is configured as a DHCP and DNS server

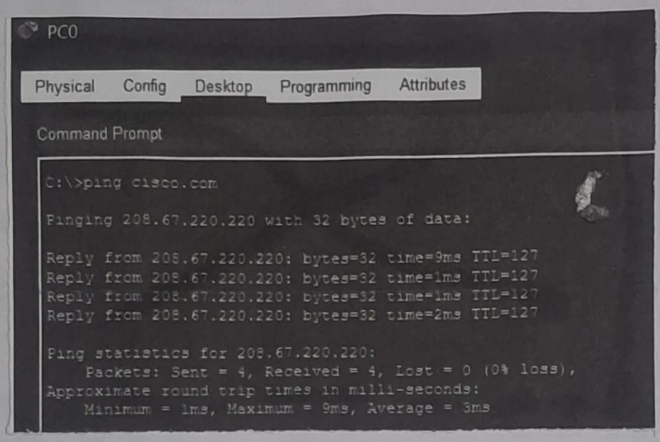
8.) The PC receives an address in the 192.168.0.x range

9.) Connectivity is verified using Ping Cisco.com


Cisco Packet Tracer - C:\Users\THARUNIKKA\OneDrive\Desktop\CLG\SEM 5\Computer Networks\Cisco\Exp 10(b).pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
PC-PT PC0
Laptop-PT Laptop0
Cloud-PT Cloud0
Server-PT Server0


No association with access point
Infrastructure Mode


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 208.67.220.220
Pinging 208.67.220.220 with 32 bytes of data:
Request timed out.
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Ping statistics for 208.67.220.220:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\>



Result :- Thus, the wireless Router, DHCP server and Internet cloud were successfully configured, and connectivity to the Cisco.com server was verified